

An explicit proof of the existence of a proper Euler magic matrix of order seven

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Abstract

We give an explicit integer matrix of order seven whose rows are orthogonal with common squared norm 4793786401980625, whose two diagonal squared sums equal this norm, and whose 49 entry squares are pairwise distinct. A rational construction combines a Householder reflection with rotations on three mirror pairs. A cyclic difference identity enforces the two diagonal conditions. The existence proof is certified directly by finite integer arithmetic on the displayed matrix. No minimality or assertion for other orders is made.

1 Statement and context

For a real matrix $X = (x_{ij})$ of order seven, write

$$D(X) = \sum_{i=1}^7 x_{ii}^2, \quad A(X) = \sum_{i=1}^7 x_{i,8-i}^2.$$

An integer matrix M is an Euler magic matrix if $MM^\top = \gamma I_7$ and $D(M) = A(M) = \gamma$ for some $\gamma > 0$. It is proper when all its entry squares are distinct. Negative entries are permitted. These conventions are the integer, order-seven specialization of Müller [2, Definition 1.1] and Kominers [1, Definition 1.1]. In particular, two zero entries would have equal squares, so properness allows at most one zero entry. Müller gives a proper order-eight construction and proves nonexistence in order three over $\mathbb{Q}$ [2, Theorems 1.3 and 1.2]; Kominers gives a proper order-five construction using mirror-coordinate rotations [1, Section 3 and Theorem 4.1]. These works provide the context for the present order-seven construction. We make no claim of priority or of an exhaustive account of the literature.

Theorem 1.

$$\exists M = (m_{ij}) \in \mathbb{Z}^{7 \times 7} \exists \gamma \in \mathbb{Z}_{>0} : \quad MM^\top = \gamma I_7, \quad \sum_{i=1}^7 m_{ii}^2 = \sum_{i=1}^7 m_{i,8-i}^2 = \gamma, \quad m_{ij}^2 \neq m_{kl}^2 \text{ for all } (i,j) \neq (k,l).$$

In this formula both cell indices range over $\{1, \dots, 7\}^2$. Thus there exists a proper Euler magic matrix of order seven. The center entry contributes once to each of D and A . The row norm equation makes γ integral even if its domain is initially taken to be the positive real numbers.

2 An explicit certificate

Put $c = 69\,237\,175$ and $\gamma = c^2 = 4\,793\,786\,401\,980\,625$. The following matrix is our witness:

$$M = \begin{pmatrix} 4535090 & 35404523 & -18199486 & 3956410 & -48268202 & 28849714 & 3277090 \\ -14516950 & -2157470 & 44513035 & -8145550 & -29803130 & -11315710 & -38914750 \\ 50566075 & 9366490 & -16347710 & -12296950 & -5931470 & -35653930 & -20598050 \\ -22675450 & -5239570 & -26298490 & 49455125 & -9541930 & -27481010 & -16385450 \\ -6634250 & -43474270 & -33406190 & -25128550 & -12120830 & 19257515 & -24383950 \\ -20966050 & -7088830 & -6064670 & -26763950 & -26709190 & -37180190 & 38146075 \\ -31745630 & 38473414 & -19223498 & -27694870 & 24273739 & -7335398 & -22939630 \end{pmatrix}. \quad (1)$$

Proof of Theorem 1. All displayed entries are integers and $\gamma > 0$. Direct integer dot products give

$$\left(\sum_{k=1}^7 m_{ik} m_{jk} \right)_{i,j=1}^7 = 4\,793\,786\,401\,980\,625 \, I_7, \quad D(M) = A(M) = 4\,793\,786\,401\,980\,625.$$

For completeness, a short exact verification of all these finite arithmetic assertions is supplied below. Sorting the 49 squared entries into $s_1 \leq \dots \leq s_{49}$ gives

$$\min_{1 \leq k < 49} (s_{k+1} - s_k) = 1\,235\,349\,458\,400 > 0.$$

Therefore every pair of distinct cells has different squares. These identities and the strict inequality verify all the required conditions. $\square$

The Gram identity also implies $M^\top M = \gamma I_7$, since M is invertible. Thus the entry squares have sum γ along every row and column as well as along both specified diagonals.

3 Rational construction

Here is a derivation of the witness, independent of the direct certificate proof. Let J be the all-ones matrix of order seven and let P be given by $P_{i,\sigma(i)} = 1$ for $\sigma = (1\,2\,3)(4)(5\,6\,7)$, with all other entries zero. Set

$$H = I_7 - \frac{2}{7}J, \quad B = PH = P - \frac{2}{7}J.$$

Since $J^2 = 7J$, $H^\top = H$ and $H^2 = I_7$. Hence B is orthogonal.

Explicit rational matrix certificates also arise in a recent rank-(2,2) Hadamard-factorization construction [4]. At a more general methodological level, symmetry-reduced exact finite certificates have been used for restricted discrete-geometric models [5]. Those results concern different matrix and geometric problems; they do not imply the Euler magic construction given here. The only permutation entry on either specified diagonal is at (4,4). Each diagonal of B therefore has one entry $5/7$ and six entries $-2/7$, so $D(B) = A(B) = 1$.

On the ordered mirror pairs (1,7), (2,6), (3,5), let L act by

$$G(\alpha_i) = \begin{pmatrix} \cos \alpha_i & -\sin \alpha_i \\ \sin \alpha_i & \cos \alpha_i \end{pmatrix},$$

and let R act by $G(\beta_i)$, where

$$\beta_1 = \alpha_2 - \alpha_3, \quad \beta_2 = \alpha_3 - \alpha_1, \quad \beta_3 = \alpha_1 - \alpha_2.$$

Both matrices fix coordinate 4. Set $Q = LBR$.

Lemma 1. *For every real triple $(\alpha_1, \alpha_2, \alpha_3)$, the matrix Q is orthogonal and $D(Q) = A(Q) = 1$.*

Proof. Orthogonality follows by multiplication. The sum $D + A$ counts the squared Frobenius norm of each of the three matching mirror blocks and counts the center twice. Left and right orthogonal transformations within a mirror pair preserve its matching block norm, while leaving the center fixed. Consequently $D(Q) + A(Q) = 2$.

Each matching block of P is zero. With $u = (1, 1)^\top$ the corresponding block of B is $-(2/7)uu^\top$, and that of Q is $-(2/7)(G(\alpha_i)u)(u^\top G(\beta_i))$. For a rank-one block $vw^\top$, its diagonal squared sum minus its antidiagonal squared sum is $(v_1^2 - v_2^2)(w_1^2 - w_2^2)$. Here the two factors before scaling are respectively $-2\sin(2\alpha_i)$ and $2\sin(2\beta_i)$. The center cancels, giving

$$D(Q) - A(Q) = -\frac{16}{49} \sum_{i=1}^3 \sin(2\alpha_i) \sin(2\beta_i) = 0.$$

Indeed, with $S_i = \sin(2\alpha_i)$ and $C_i = \cos(2\alpha_i)$ the sum is

$$S_1(S_2C_3 - C_2S_3) + S_2(S_3C_1 - C_3S_1) + S_3(S_1C_2 - C_1S_2),$$

whose six terms cancel in pairs. Together with $D + A = 2$, this proves the claim. $\square$

Choose $\tan(\alpha_i/2) = 1/2, 1/4, 1/6$, respectively. The exact left cosine–sine pairs are

$$(3/5, 4/5), \quad (15/17, 8/17), \quad (35/37, 12/37).$$

The difference formulas give the right pairs

$$(621/629, 100/629), \quad (153/185, -104/185), \quad (77/85, 36/85).$$

All entries of Q are rational and their denominators divide $c = 7(5 \cdot 17 \cdot 37)^2$. Multiplying out $cLBR$ gives exactly (1). This is an exact rational specialization, with no trigonometric approximation. Properness is established for this specialization by the certificate, not for arbitrary angle triples: at the zero triple, $Q = B$ has repeated squared entries.

4 Exact certification and reproducibility

The displayed matrix alone is sufficient input for the following Python 3 check. Python integers have arbitrary precision [3]; no external libraries, floating-point tolerance, or search output are used. Assign to M the seven rows in (1) as lists of integers, then execute:

```
c = 69237175
gamma = 4793786401980625
assert gamma == c*c and gamma > 0
assert len(M) == 7 and all(len(row) == 7 for row in M)
assert all(type(x) is int for row in M for x in row)
for i in range(7):
    for j in range(7):
        dot = sum(M[i][k]*M[j][k] for k in range(7))
        assert dot == (gamma if i == j else 0)
assert sum(M[i][i]**2 for i in range(7)) == gamma
assert sum(M[i][6-i]**2 for i in range(7)) == gamma
s = sorted(x*x for row in M for x in row)
assert len(s) == 49
assert min(b-a for a, b in zip(s, s[1:])) == 1235349458400
```

The accompanying `verify.py` includes the complete integer input. A separate verification implementation included in the ancillary research archive accumulated column outer products, tested all $\binom{49}{2} = 1176$ unordered pairs, and reconstructed Q using exact rational arithmetic without importing the generator. It agreed with the displayed certificate. These are finite exact checks, not an exhaustive search over matrices. The proof can also be checked by ordinary integer arithmetic without relying on Python.

For normalization, determinants give $(\det M)^2 = \gamma^7$. Prime valuations show that any positive integral γ in an order-seven integer witness is a square. Dividing by its positive square root gives a rational orthogonal matrix; negating one row if needed makes its determinant positive without changing any squared condition. Conversely, clearing a common denominator of a rational witness gives an integer witness. No determinant sign restriction is part of Theorem 1.

5 Scope, limitations, and assistance disclosure

The result establishes existence in order seven only. It gives no minimum height or minimum value of γ , no classification, no universal properness of the rotation family, and no existence statement for larger odd orders. It imposes neither positivity of entries nor primitivity. Arbitrary row or column permutations need not preserve the two specified diagonals. Earlier exploratory numerical searches, restricted parameter searches, and obstruction calculations are not premises of this proof, and no completeness assertion about those searches is made. The literature context above does not establish a historical priority claim.

Computational and language-model assistance were used in the research workflow for exploring constructions, producing and checking code, developing proof arguments, and preparing this manuscript. Discovery included numerical exploration and symbolic computation; the rational generator used Python and SymPy. Certification used exact integers and rational numbers and was repeated in a separate language-model referee context. That internal check is not journal peer review. No proof-assistant formalization is claimed. The mathematical justification is the explicit witness and the elementary identities and exact checks provided here; model confidence is not evidence.

References

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